PUSHOUTS OF SCHEMES ALONG CLOSED IMMERSIONS EXIST, ARE FIBRE SQUARES BY CLOSED IMMERSIONS, AND CARRY THE STRUCTURE-SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19986-20004 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20006-20014, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Foundational case of the chapter's pushout series; its companion properties-above transfer law is the next candidate in this same wave (sequenced L16145). The proof is a verification of a specialization's hypotheses — an explicit check, not a bare pointer.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's proposition pushout-along-closed-immersion-and-integral (the general pushout engine for closed-immersion-plus-integral spans) and its Situation setup.

Source reference. Original lemma 0E25: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19758–19761. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19716. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : Z \to X$ and $j : Z \to Y$ be closed immersions of schemes. Then the pushout $Y \amalg_Z X$ exists in the category of schemes. Picture*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle a} \\ Y & \xrightarrow{b} & Y \amalg_Z X \end{array}$$

The diagram is a fibre square, the morphisms a and b are closed immersions, and there is a short exact sequence

$$0 \to \mathcal{O}_{Y \amalg_Z X} \to a_*\mathcal{O}_X \oplus b_*\mathcal{O}_Y \to c_*\mathcal{O}_Z \to 0$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

where $c = a \circ i = b \circ j$.

Proof. This is a special case of Proposition 0E25. Observe that hypothesis (3) in Situation 0ECI is immediate because the fibres of j are singletons. Finally, reverse the roles of the arrows to conclude that both a and b are closed immersions. $\square$